

Project Milestone 2 – Final Submission Draft

Multi-Agent Evacuation Under Dynamic Hazards

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1 Progress Overview

Since Milestone 1, the project has progressed from a conceptual framework to a working simulation and evaluation pipeline.

We have implemented the evacuation environment and conducted initial experiments across multiple disaster scenarios using baseline policies.

Preliminary results have been obtained, showing differences in performance under varying hazard conditions.

2 Methodology Development

The evacuation problem is modeled as a partially observable decision process on a campus road network under dynamic hazard conditions. Agents receive local and potentially noisy observations about the environment and must make sequential routing decisions toward shelters as environmental conditions evolve. Our approach is inspired by graph-based exploration methods such as BEAG [1], which leverage adaptive grid structures and breadth-first exploration to improve efficiency in sparse reward environments.

2.1 Graph Environment

The campus road network is extracted from OpenStreetMap using OSMnx and represented as a directed graph $G = (V, E)$. Each node $v \in V$ corresponds to a road intersection or pedestrian waypoint. Each edge $e = (u, v) \in E$ represents a traversable road segment, with a dynamic edge weight defined as:

$$w(e, t) = d(u, v) \cdot (1 + \alpha \cdot \text{slope}(e) + \beta \cdot h_{\text{snow}}(e, t))$$

where $d(u, v)$ is the Euclidean distance between endpoints, $\text{slope}(e)$ is the terrain gradient derived from USGS digital elevation models, $h_{\text{snow}}(e, t)$ is the snow depth on edge e at time step t , and α, β are tunable hyperparameters controlling the relative penalty of terrain and snow. When $h_{\text{snow}}(e, t)$ exceeds a predefined blockage threshold τ , the edge is marked as impassable and removed from the routing graph for all subsequent agents.

Both walking and driving mobility are supported. Pedestrian agents traverse the full pedestrian network, while vehicle agents are restricted to drivable road segments. Shuttle vehicles follow fixed route corridors with designated stop nodes.

2.2 Hazard Process

Snow accumulation evolves stochastically over time. At each simulation step, the snow depth on each edge increases according to:

$$h_{\text{snow}}(e, t + 1) = h_{\text{snow}}(e, t) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(\mu_s, \sigma_s^2)$$

The accumulation rate μ_s and variance σ_s^2 are set per scenario. The blizzard scenario uses a higher μ_s , causing edges to fail progressively over time and forcing agents to adapt their routes dynamically. In the earthquake scenario, road blockages are instead sampled from a Bernoulli process at each step, reflecting sudden structural failures. The compound disaster scenario combines both processes simultaneously.

2.3 Agent Observation Space

Because the environment is partially observable, no agent has access to the full global state. At each time step t , each agent receives a local observation vector $o_t \in \mathbb{R}^d$ comprising:

- The current node identifier (one-hot encoded over $|V|$ nodes)
- Binary traversability signals for each adjacent edge, corrupted by Gaussian noise
- A direction vector toward the nearest unblocked shelter
- The agent’s accumulated hazard exposure up to time t
- Agent type encoding (pedestrian, car, or shuttle)

The noise level applied to edge traversability signals is parameterized by scenario severity, reflecting the reduced observability during whiteout or post-seismic conditions.

2.4 Baseline Policies

Three heuristic policies serve as benchmarks.

Round Robin. Agents are assigned to shelters sequentially by agent identifier, without regard to distance, congestion, or hazard state. This policy serves as a naive lower-bound reference.

Nearest Shelter. Each agent is assigned to the closest shelter by current network distance, recomputed at each step to account for edge blockages. This represents a strong practical baseline consistent with conventional evacuation guidance.

2.5 DRQN Architecture

Since the true environment state is not directly observable, we train routing policies using Deep Recurrent Q-Networks (DRQN) [2]. The network processes a sequence of $L = 8$ consecutive observations $\{o_{t-L+1}, \dots, o_t\}$ through the following architecture:

- **Encoder:** A fully connected layer mapping the observation $o_t \in \mathbb{R}^d$ to a 128-dimensional embedding with ReLU activation.
- **LSTM:** A single-layer LSTM with hidden size 128, maintaining a recurrent belief state across time steps to compensate for partial observability.
- **Output head:** A linear layer mapping the LSTM hidden state to Q-values over the action space A .

The action space A consists of moving to an adjacent graph node, waiting in place, or requesting shuttle pickup at the nearest stop. Training uses Double DQN target updates to reduce overestimation bias. A sequence replay buffer stores fixed-length episode fragments of length L , and a burn-in period of 4 steps is applied during hidden state initialization to reduce gradient corruption from stale recurrent states. A best-checkpoint tracking mechanism retains the model weights corresponding to the highest validation reward observed during training.

3 Hazard Scenarios and Operational Conditions

The project currently evaluates four disaster scenario types. Each scenario represents a different level of operational stress during a campus evacuation.

3.1 Baseline Scenario

The baseline scenario represents normal to mildly disrupted conditions. Snow accumulation and road blockage effects are limited compared to other scenarios. This scenario is used as a reference case to measure policy performance under relatively stable conditions.

3.2 Blizzard Scenario

The blizzard scenario represents severe winter weather with dynamic snow accumulation. Snow depth increases over time, and roads may become blocked once snow exceeds a predefined threshold. Observability is reduced, meaning agents make decisions based on noisier and less reliable local information. Route reliability degrades over time, making early evacuation and adaptive routing strategies increasingly important.

3.3 Earthquake Scenario

The earthquake scenario models structural disruptions and high uncertainty in path availability. Road blockages occur more aggressively and route accessibility may change suddenly due to damaged infrastructure. In this setting, shortest routes may fail unexpectedly, making adaptive replanning critical.

3.4 Compound Disaster Scenario

The compound disaster scenario combines multiple hazard processes, such as structural disruptions together with snow accumulation and reduced visibility. This scenario represents the highest operational stress in the current study. Accessibility and environmental exposure deteriorate simultaneously, typically resulting in lower reachability and increased risk.

3.5 Experimental Usage of Scenarios

All policies are evaluated across the same four scenarios to ensure fair comparisons. The same evaluation framework and key performance indicators (KPIs) are applied in each scenario. This design allows the experiments to measure both policy effectiveness and policy robustness under increasing hazard severity.

4 Evaluation Metrics

Several key performance indicators are used to evaluate evacuation performance.

Reached Rate. The fraction of agents that successfully reach a shelter. This metric represents the core success rate of the evacuation system.

Alive Rate. The survival ratio of agents under hazard exposure. This metric serves as a safety upper bound in the current simulation setup.

Average Exposure. The accumulated hazard exposure per agent during evacuation. Lower values indicate safer evacuation trajectories.

Evacuation Time (T90 and T95). The number of simulation steps required to evacuate 90% and 95% of agents. These metrics measure evacuation efficiency and system throughput.

Fairness Metrics. Population groups are evaluated separately using faculty reached rate and staff reached rate. The difference between the two groups (reach rate gap) is used to quantify fairness.

Network Bottlenecks. The most congested network edges are recorded to identify operational bottlenecks and potential infrastructure vulnerabilities.

5 Implementation Progress

The current system architecture consists of the following key components:

5.1 Layer 1: LLM-Based Human Behavior Profiling

We utilized the **Groq** platform to support an LLM-based persona generation pipeline, producing 20 distinct behavioral profiles across four categories: students, faculty, staff, and visitors.

- **Heterogeneous Parameters:** Quantitative attributes including speed, compliance, panic levels, and observation errors are directly incorporated into the simulation.
- **Behavioral Diversity:** This profiling framework enables the simulation of heterogeneous human responses to stress, uncertainty, and imperfect environmental perception.

5.2 Layer 2: Zone-Level Coordination and Route Recommendation

In addition to individual-level navigation, we extended the system with a zone-level planning and recommendation layer. This component converts learned routing behavior into actionable evacuation guidance at both the individual and group levels.

- **Route Recommendation Prototype:** The system can generate a recommended shelter, complete node path, traversed edges, evacuation steps, hazard exposure, and replanning history for a given starting location.
- **Zone-Based Assignment:** We implemented a zone-level shelter assignment module that groups evacuee origins into zones and generates primary and backup shelter recommendations.
- **Capacity-Aware Coordination:** The latest version incorporates shelter capacity constraints, demand balancing, and route-feasibility screening, moving beyond purely nearest-shelter assignment toward a more operational decision-support framework.

5.3 Layer 3: Advanced DRQN Navigation

The navigation model has been upgraded to a **Dueling DQN + GRU** architecture.

- **Top-k Neighbor Selection:** The agent evaluates the top- k adjacent nodes on the OpenStreetMap (OSM) graph, allowing more flexible branching under partial observability.
- **Curriculum Learning:** We implemented a multi-stage training pipeline, from easier scenarios to progressively more severe conditions, to improve stability in blizzard and earthquake environments.

6 Preliminary Results

6.1 Summary of Results

Table 1 summarizes the reached rate and average hazard exposure for the two primary baselines across all four scenarios.

Table 1: Baseline policy comparison across scenarios (reached rate and average exposure).

Scenario	Reached Rate		Avg. Exposure	
	Round Robin	Nearest	Round Robin	Nearest
Baseline	0.2654	0.2909	294.88	273.38
Blizzard	0.0061	0.0788	203.03	195.30
Earthquake	0.0073	0.0427	390.82	378.45
Compound Disaster	0.0055	0.0291	412.54	403.57

6.2 Analysis

Across all four scenarios, the nearest shelter policy consistently outperforms round robin on both reached rate and average exposure, confirming that distance-aware routing provides a meaningful improvement over naive assignment. However, the magnitude and pattern of these gains reveal several important observations.

Reached rate degrades sharply under hazard. Under the baseline scenario, the nearest shelter policy achieves a reached rate of 0.29. This drops to 0.08 under blizzard conditions and further to 0.04 under earthquake and compound disaster scenarios. The sharp degradation under dynamic hazards indicates that static distance-based routing is poorly suited to environments where the network topology changes over time. When edges fail mid-evacuation, agents assigned to now-unreachable shelters have no mechanism to replan, leading to high failure rates.

The blizzard scenario reveals the cost of delayed routing. The relatively low reached rate under blizzard conditions (0.08 for nearest shelter) is consistent with the progressive nature of snow accumulation: edges that are traversable at the start of evacuation may become blocked before agents arrive. This motivates the need for anticipatory routing policies that account for future hazard evolution, rather than greedy shortest-path assignment based on current network state.

The earthquake scenario produces higher exposure despite lower reached rates. Average exposure under earthquake conditions (378 for nearest) is substantially higher than under blizzard (195), even though the earthquake reached rate is lower. This suggests that in earthquake scenarios, agents spend more time navigating before failing to reach shelter, likely due to sudden mid-route blockages forcing longer detours. In contrast, blizzard conditions tend to strand agents earlier, resulting in shorter but ultimately unsuccessful trajectories.

Compound disaster represents a floor on current policy performance. The compound scenario combines both hazard processes and produces the lowest reached rate (0.03) and highest average exposure (403) across all conditions. The marginal improvement of nearest over round robin narrows under compound conditions, suggesting that distance-aware heuristics provide diminishing returns as the environment becomes increasingly unpredictable. This finding motivates the use of a learning-based policy, such as DRQN, which can adapt to dynamic and partially observable conditions through recurrent memory and online policy improvement.

Overall baseline performance remains low. Even under the mildest conditions, the reached rate does not exceed 0.30 for either baseline. This is likely attributable to network capacity constraints: when a large number of agents simultaneously route toward nearby shelters, congestion at high-demand nodes increases traversal time and exposure. Future work will investigate congestion-aware routing and load-balancing mechanisms as a complement to learning-based policies.

These results are consistent with expectations, as distance based routing improves performance compared to naive strategies, but fails under dynamic and uncertain conditions.

7 Evaluation and Results

7.1 DRQN vs. Baseline Policies

The DRQN policy was evaluated against *Round Robin* and *Nearest Shelter* heuristics in the `enterprise_baseline` scenario.

Table 2: Performance Comparison across Policies

Policy	Avg. Reached Rate	Avg. Total Exposure	Improvement
Round Robin	0.2654	286.14	-
Nearest Shelter	0.2909	246.92	-
DRQN (Ours)	0.9627	5.41	+262.7%

7.2 Analysis of Findings

- **Hazard Resilience:** Under extreme blizzard conditions, DRQN maintains a reached rate of approximately 0.716, whereas baseline policies collapse to near-zero performance.
- **Behavioral Impact:** The introduction of LLM-profiled heterogeneous personas resulted in a measurable decrease in reached rate compared to uniform-agent settings, confirming that population heterogeneity significantly affects evacuation efficiency.
- **Practical Implication:** These results suggest that learning-based routing is substantially more robust than static heuristic assignment in dynamic and partially observable environments.

7.3 Robustness Under Hazard Severity

To further evaluate robustness, we conducted additional experiments across progressively harsher blizzard settings. During this process, we identified an initialization issue in which some agents were spawned at poorly connected or effectively isolated locations under high blockage conditions. After correcting the spawn logic and calibrating the initial blockage setting, performance improved substantially in the most severe scenarios. This debugging process strengthened the reliability of

the evaluation pipeline and confirmed that the learned policy remains effective under severe hazard conditions when evaluation artifacts are controlled.

7.4 Scaling Performance

We also evaluated the learned policy under larger evacuation populations. The scaling experiments increased the number of evacuees well beyond the initial small-scale setting and showed that the DRQN policy maintained consistently strong reached rates under the current simulation assumptions. These results suggest that the model is not limited to a narrow low-density setting. At the same time, they indicate that future stress tests should place greater emphasis on shelter capacity, congestion, and resource competition, since agent count alone may not fully reveal system bottlenecks.

8 Future Work

The next phase of the project will extend the current framework from policy-level evaluation toward population-aware evacuation simulation and LLM-assisted decision support under dynamic hazards.

Planned work includes:

- Constructing a population simulation that reflects the current demographic composition of the University of Utah community, including groups such as students, faculty, staff, and visitors.
- Distributing these heterogeneous agents across realistic campus locations based on their likely spatial presence and activity patterns, in order to better approximate real-world evacuation conditions.
- Evaluating how different population groups respond to disaster scenarios when exposed to dynamic hazards, limited observability, and changing route accessibility.
- Extending the current evacuation framework to analyze group-specific differences in routing behavior, hazard exposure, congestion, and shelter reachability.
- Incorporating LLM-assisted route recommendation to provide adaptive evacuation suggestions based on agent type, environmental conditions, and available shelter options.
- Producing a final reproducible comparison between heuristic routing, learning-based routing, and LLM-assisted recommendation strategies in a realistic campus population setting.

These extensions will allow the project to move beyond synthetic population settings and toward a more realistic simulation of evacuation behavior on the University of Utah campus. They will also help evaluate whether LLM-assisted guidance can improve evacuation decision-making for heterogeneous populations under disaster conditions.

References

- [1] Y. Yoon, G. Lee, S. Ahn, and J. Ok, “Breadth-first exploration on adaptive grid for reinforcement learning,” in *Proceedings of the International Conference on Machine Learning (ICML)*, 2024.
- [2] M. Hausknecht and P. Stone, “Deep recurrent Q-learning for partially observable MDPs,” *arXiv preprint arXiv:1507.06527*, 2015.